UNIVERSITY OF UYO

DEPARTMENT OF MATHEMATICS

FIRST SEMESTER EXAMINATION 2022/2023 SESSION

MTH 412: PARTIAL DIFFERENTIAL EQUATIONS TIME: 2 HOURS

INSTRUCTIONS: ATTEMPT QUESTION 1 AND ANY OTHER 3 QUESTIONS ONLY

In the equations, take $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$

1.

a) Integrate $ydx + xdy + 2zdz = 0$

b) Solve the differential equation $xdx + ydy = zdz$

c) $yzdx + 2xzdy - 3xydz = 0$

2. Obtain the integral surface of the nonlinear partial differential equation $pq = xy$ which passes through the curve $z = x,\ y = 0$.

3. Show that the equations $xp = yq, z(xp + yq) = 2xy$ are compatible and solve them.

4. Find the integral curves of the following systems:

a) $\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}$

b) $\frac{dx}{yz} = \frac{dy}{xz} = \frac{dz}{xy}$

c) $\frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)}$

5. Find the general integral of the following linear partial differential equations:

a) $xp + yq = z$

b) $z(xp - yq) = y^2 - x^2$

c) $yzp + zxq = qxy$

6. Find the integral surface of the linear partial differential equation

$$x(y^2 + z)p - y(x^2 + z)q = (x^2 - y^2)z$$

which contains the straight lines $x + y = 0$ and $z = 1$.